ENERGY & GRID HARDENING · INNOVATELA DUAL-USE HACKATHON · USC, OCTOBER 2026

POWERUP

A startup that owns a fleet of Base batteries, with a disaster layer for it: **PowerUP plans** each home's backup days before a wildfire shutoff, **Interlock guards** every order in seconds, and neighbors share one **island** behind SCE's switch.

Automation that knows when to watch, ask, slow, hold and hand to a person. We built that ladder twice: over days, and in seconds.

Every number on this poster is labelled real, assumed or simulated. Reproduce the results with **node pilot/app/cli.mjs**.

THE REAL SHUTOFF, MALIBU, OCTOBER 2026

6,874
customers on six SCE circuits lose power to prevent fire, Oct 9 to 18 (real)

Oct 15
the Bryson Fire starts at 4:41 PM, during the shutoff (real)

6.5 days
from the first Santa Ana wind signal (Oct 9, 2:14 AM) to the power going off (real)

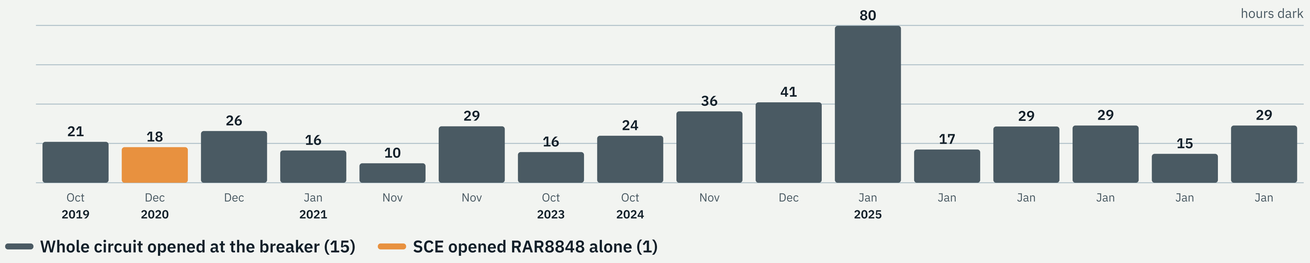
69,746
SCE customers out at 7 AM on Oct 16 (real)

A battery fleet fails in two ways. It is not ready: half empty when the power goes off, with Medical Baseline homes treated like everyone else. Or its own software is the disaster: a sell-down three hours before the shutoff, a stolen key, a replayed order, export onto a line crews think is dead, or a recharge rush that overloads the circuit when it comes back.

The pilot: 456 homes behind SCE switch RAR8848 on the Cuthbert circuit, Latigo substation, 9 km from the Bryson burn.

189 of them have a battery: our **160 free Base Cores** (39.2 kWh, 20 kW) plus 29 of the homes' own. 16 are Medical Baseline homes (assumed share). Upstream, a 66 kV line into Latigo runs about 650 m inside the Bryson perimeter (real, route approximate).

EVERY SHUTOFF THAT CUT THESE 456 HOMES, 2019 TO 2025 (REAL SCE RECORDS): 16 TIMES, 435 HOURS



PowerUP plans, over days, on the real timeline

Starts from SCE's own list of circuits it may cut.

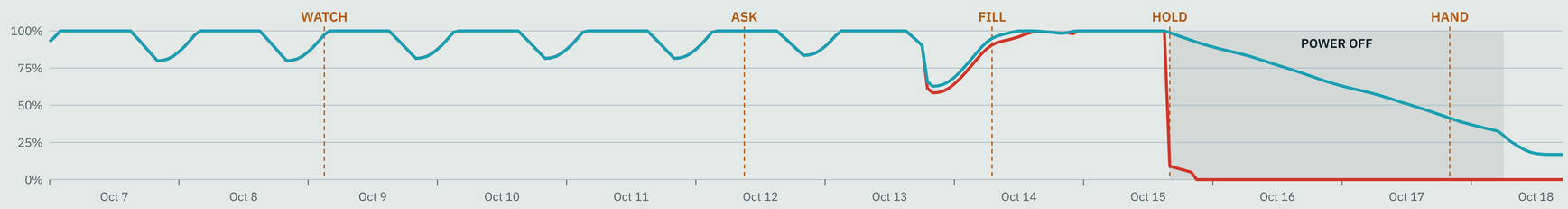
FRI OCT 9, 3 AM — WATCH
First wind warning. Medical Baseline homes fill to 100% and hold.

MON OCT 12, 9 AM — ASK
SCE names Cuthbert. Owners are asked to pre-fill; everyone else keeps a 50% floor.

WED OCT 14, 7 AM — FILL
The shutoff is sure. Every battery whose owner said yes fills, under the circuit's headroom. Nothing is sold.

THU OCT 15, 4 PM — HOLD
Power off. Every battery islands, never exports, and runs its home on a budget: 8, 16 or 32 kWh a day.

SAT OCT 17, 8 PM — HAND
Winds stop. PowerUP proposes a staggered recharge, one circuit at a time; SCE's shutoff officer approves.



FLEET CHARGE, 189 BATTERIES, OCT 7 TO 18 (SIMULATED ON REAL DATA)

Guard on

Guard off

With the guard off, a market bid sells the fleet down to 9% three hours before the shutoff, and every battery home is dark by midnight. With it on, the fleet enters the shutoff at 99% and carries every home to the restart.

Interlock guards, in seconds

Every order to the fleet, from any sender, passes eight checks in order. Each check answers on the gentlest rung that keeps homes and the circuit safe. **It can only shrink, slow or stop an order, never create one.** PowerUP's own orders pass through it too, and the reserve PowerUP plans for each home becomes the floor Interlock keeps against every sender: our daily optimizer, a market bid and SCE's emergency event.

FIVE RUNGS, GENTLEST FIRST

RUN — The order passes as sent.
CLAMP — Slowed or capped to the safe part.
CANARY — 1% of homes, then 10%, then all.
HOLD — Nothing runs until it is safe.
HAND — A person decides.

Orders are shaped like OpenADR 3 events. Policy packs change the limits per region without changing the engine.

1 SCHEMA VALIDATION
A malformed order never reaches a battery.

2 AUTHENTICATION
Signed by a sender on the trusted list, or held for a person.

3 REPLAY PROTECTION
Issued in the last 15 minutes and never seen before.

4 PRIMACY
Safety and PowerUP's plan outrank trading and market bids.

5 BACKUP RESERVE
Never below 20%, 50% for Medical Baseline, or the reserve PowerUP planned.

6 RATED POWER
Within the 20 kW inverter and the room left in each battery.

7 HOSTING CAPACITY
No export onto a de-energized line; no charging over the circuit's limit.

8 STAGED ROLLOUT
New fleet-wide orders go out to 1%, then 10%, then everyone.

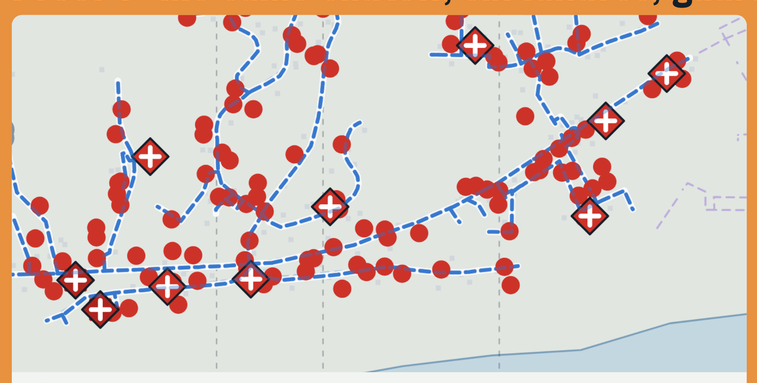
A SILENT BATTERY IS NEVER COUNTED AS POWER

Interlock tells why a battery went quiet by what went quiet with it. Before the cell towers fail, PowerUP sends each battery its full plan, so a silent battery keeps following it alone, and homes report over mesh radio to a Starlink dish.

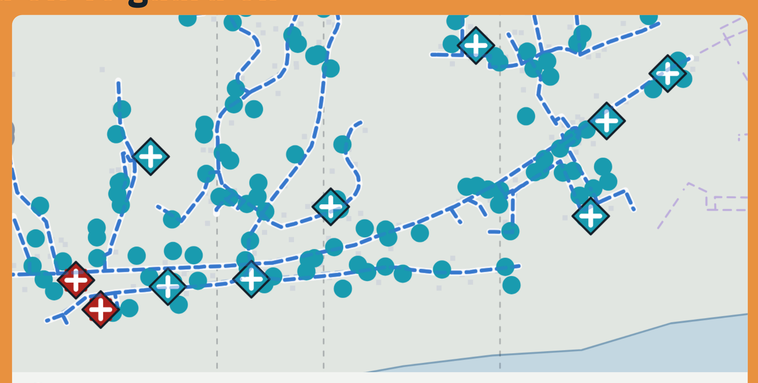
TRIPPED — Its whole circuit went quiet together.
NOT TALKING — Its cell tower or internet went quiet. It follows its plan alone.
BROKEN — Quiet alone, after a fault code: to a person.
UNSURE — Anything else: ask the owner first.

Proof 1: the same shutoff, six failures, guard off vs guard on

RAR8848 PILOT · OCT 7 TO 18, 2026



Guard off, a day into the shutoff: every battery is empty



Guard on, the same hour: every battery home has power

Battery homes that go dark: **189 → 0**

Medical Baseline homes that go dark: **16 → 0**

Fleet charge when the power goes off: **9% → 99%**

Battery-hours pushed onto a dead line: **567 → 0**

Overload over Cuthbert's cap at the restart: **3.4 MW → 0**

Home-hours kept powered by batteries: **945 → 11,596**

WHEN (2026)	INJECTED FAILURE (SIMULATED)	INTERLOCK	WHAT THE CHECK FOUND	GUARD OFF
Tue Oct 13, 5 PM	SCE emergency event: every home exports 5 kW for 2 hours	CLAMP	Slowed on 34 homes so no backup drops below its floor	189 homes drained
Wed Oct 14, 5 PM	Replay: a captured Oct 8 discharge order is sent again	HOLD	Issued 8,700 minutes ago, id already seen	130 kWh drained
Thu Oct 15, 1 PM	Market bid: sell every battery down to 10% by 4 PM	CLAMP	PowerUP's Fill outranks it; target raised to the floor	6,668 kWh sold
Thu Oct 15, 8 PM	Stolen key posing as SCE: discharge every battery at 20 kW	HOLD · HAND	Signer not on the trusted list; a person reviews it	189 batt-hours on a dead line
Fri Oct 16, noon	Market export onto the de-energized Cuthbert line	HOLD	Circuit open: homes island and serve themselves	378 batt-hours on a dead line
Sun Oct 18, 6 AM	Optimizer rebound: recharge everything at 20 kW when power returns	CLAMP	3.83 MW against a 0.97 MW limit: all slowed to 11%	3.4 MW over the cap

PowerUP's own five orders: 5 of 5 pass, none held or clamped. The same run on all six Malibu circuits (1,032 battery homes): 1,028 dark with the guard off, none with it on. At Malibu's real battery share today, 7.1 per 100 homes (SCE interconnection records): 483 of 484 dark with it off, none with it on.

PROOF 2: THE EVENING BEFORE

WED OCT 14, 2026: AN ORDINARY TRADING EVENING, INTERLOCK ALONE

The same 189 battery homes, Cuthbert's real October load shape scaled to the branch (assumed), SCE time-of-use prices (assumed), one SCE event and eleven failures (simulated): a stolen key, a replay, a price glitch, a bad forecast, a market export while SCE has RAR8848 open, and more.

GUARD	OFF	ON
Homes drained below their backup	189	**0**
Export onto the opened line, battery-minutes	2,385	**0**
Over the branch's 0.97 MW cap, MW-minutes	306	**0**
Revenue that evening	$2,795	**$3,331**

Safe and still earning: the guarded fleet makes 1.19 times what the unguarded one does. Interlock doubles as our test bench: replay a new dispatch policy, then roll it out to 1%, 10% and 100% of the fleet.

ISLANDS BEHIND SCE'S SWITCH

SCE already opens partial circuits: 13 of the 55 shutoffs on these six circuits since 2019 were opened at a switch on the line (real). The island planner finds homes that go dark only because their feed runs through a risky stretch, and plans an opt-in island behind the existing switch, with SCE opening and closing it. Three hard rules:

1. Never re-energize overhead line in a fire area.
2. A switch between the island and every dead neighbor.
3. At least one grid-forming source on the island.

STATED ASSUMPTIONS, TAKEN AS GIVEN

SCE can open RAR8848 and run the 456 homes behind it as an island. Our batteries are grid-forming: they set the island's voltage and frequency, not only follow it.

The pilot carries all 16 real shutoffs for every subscriber at the critical budget with clear-sky solar (150 batteries is the minimum), and 15 of 16 with smoke halving the solar (all 16 would take 228).

THE PILOT BUSINESS

160 free Base Cores in volunteer homes: Medical Baseline first, then solar roofs, which refill every day. **$9.99 a month** buys emergency power for a home with no battery (226 homes at an assumed 90% opt-in). In a shutoff, everyone on the island pays **$1.20 per kWh**, twice SCE's top time-of-use rate (assumed); SCE bills it and keeps $0.20 for delivery, sellers get $1.00. A host uses the bottom 20% of our battery first, at the same price.

INCOME PER BATTERY	LABEL	$ / YR
Trading, CAISO prices for SCE	real prices	334
Capacity payments	assumed	941
Emergency grid events	assumed count	627
Subscriptions	assumed opt-in	169
Shutoff power sales	assumed rates	90
Holding full from SCE's notice	real prices	−28

Fleet about $2.24M after the 30% tax credit (installed cost assumed), net about $341,000 a year. Payback 6.6 years with all income, 25 on trading, subscriptions and shutoff sales, 46 on trading alone.

FUNDING DOE SBIR Phase I through the Office of Electricity and CESER first; DOE GRIP with SCE as the applicant to scale; FEMA's Hazard Mitigation Grant Program after a declared disaster, with California's Microgrid Incentive Program for the shared battery. Built: the bridge reports what grant programs score: homes kept powered, Medical Baseline hours protected, overload avoided.

GLOBAL South Africa's load shedding is a planned outage, announced ahead and repeated several times a day: PowerUP's ladder running daily, with hospitals and clinics in the Medical Baseline role. Built: region policy packs swap the risk signal, the priority register and the limits without changing the engine.

RUN THE REPLAY YOURSELF

mrbese.github.io/PowerUp/

Code: github.com/mrbese/PowerUp · 60 node tests (npm test)

DATA

SCE shutoff records, circuit lines, load profiles and interconnection records; NIFC fire perimeter; California Energy Commission transmission map; CAISO day-ahead prices; NWS Los Angeles forecasts. Switch positions and transformers are estimated.

RESPONSIBLE USE

Misuse: locking owners out of their batteries, a single point of attack, or pricing power up mid-crisis. Limits: Interlock never goes below a home's own backup, PowerUP sees a priority tier, not a medical record, a person approves every restart and island, and the price is fixed in advance.